



Southeast University

Department of Computer Science & Engineering

COMPLEX ENGINEERING PROJECT (CEP)

Statistical Investigation of Online Shopping Behavior

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Section – 8

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1. Introduction

The film industry generates enormous volumes of data — from audience engagement metrics to box office revenues — that offer rich opportunities for statistical investigation. Understanding what factors drive a movie's rating is not only academically interesting but also commercially vital for studios, streaming platforms, and investors.

This report presents a comprehensive statistical analysis of the IMDb Popular Movies Dataset (enriched with TMDB data), sourced from Kaggle. The dataset contains 5,000 movie records spanning from 2000 to 2025, capturing attributes such as IMDb rating, vote count, Duration, genre, budget, and revenue. The analysis employs descriptive statistics, probability theory, Bayesian reasoning, the Central Limit Theorem (CLT), hypothesis testing, correlation analysis, and regression modeling to uncover meaningful insights about what makes a movie highly rated.

The primary focus variables are: Rating, Duration, Votes, Budget, and Revenue — with Rating as the key outcome variable for prediction.

The objectives of this project are threefold:

- Characterize the statistical distributions and central tendencies of movie-related variables.
- Apply inferential statistical methods to test hypotheses about audience behavior and movie success.
- Build and interpret regression models that predict IMDb rating.

2. Dataset Description

The dataset employed in this study is the IMDb Popular Movies with TMDB Enrichment Dataset, sourced from the Kaggle platform (<https://www.kaggle.com/datasets/areazare/imdb-popular-movies-with-tmdb-enrichment/data>). It was constructed by combining IMDb movie data with additional enrichment from TMDB (The Movie Database) and is widely used for rating prediction and entertainment analytics.

2.1 Dataset Overview

The table below summarizes the dataset used in this study:

Item	Value
Total Records	5,000
Total Variables	21
Numerical Variables	8 (Year, Duration, Rating, Votes, Budget, Revenue, Popularity, TMDB_ID)
Categorical Variables	13 (IMDb_ID, Title, Genres, Director, Writer, Actors, etc.)
Missing Values (Duration)	1
Duplicate Records	0
Total Records (After Preprocessing)	4,999
High-Rated Sessions (Rating $\geq$ 7.0)	1,826 (36.53%)

Year Range	2000 – 2025
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Table 1: Dataset Overview

2.2 Variable Descriptions

The table below summarizes the six key variables used in this study:

Variable	Type	Description
Rating	Continuous	IMDb rating score (1.0 – 9.0)
Duration	Integer	Movie duration in minutes (46 – 374 min)
Votes	Integer	Number of user votes on IMDb (23,766 – 3,026,849)
Budget	Integer	Production budget in USD (0 = not disclosed)
Revenue	Integer	Box office revenue in USD (0 = not disclosed)
Genres	Categorical	Comma-separated list of genre tags (e.g., Action, Drama, Comedy)

Table 2: Key Variables Used in This Study

2.3 Preprocessing Steps

The following preprocessing steps were applied before statistical analysis:

- Missing value analysis was performed using `isnull().sum()`, confirming that only 1 record was missing Duration and 5 were missing TMDb_ID. All critical analytical variables (Rating, Votes, Duration) were complete.
- 1 record with missing Duration was removed, yielding a clean dataset of $n = 4,999$ records.
- Budget and Revenue values of 0 were treated as undisclosed rather than true zeros, and were excluded from correlation/regression analyses involving these fields.
- A binary variable HighRated was derived: $\text{HighRated} = 1$ if $\text{Rating} \geq 7.0$, else 0. This threshold corresponds to the upper 36.53% of the rating distribution and serves as the outcome for probability and classification analyses.
- No duplicate rows were detected (0 duplicates confirmed).
- No outlier removal was performed. Extreme observations (e.g., very high Votes for blockbusters) were preserved to maintain data integrity.

2.4 Class Distribution

Class	Count	Percentage
Low/Mid Rated (Rating < 7.0)	3,173	63.47%
High Rated (Rating $\geq$ 7.0)	1,826	36.53%
Total	4,999	100.00%

Table 3: Rating Class Distribution

3. Statistical Analysis

3.1 Descriptive Statistics

Descriptive statistics quantify the central tendency, dispersion, and shape of each variable's distribution. For a dataset $X = \{x_1, x_2, \dots, x_n\}$, the following formulas apply:

3.1.1 Measures of Central Tendency

Mean: $\bar{x} = (1/n) \sum_i x_i$

Median: middle value when data is sorted (Q2)

Mode: most frequently occurring value

3.1.2 Measures of Dispersion

Sample Variance: $s^2 = (1/(n-1)) \sum_i (x_i - \bar{x})^2$

Standard Deviation: $s = \sqrt{s^2}$

Range = Max – Min

IQR = Q3 – Q1

3.1.3 Measures of Shape

Skewness = $(n/((n-1)(n-2))) \sum_i ((x_i - \bar{x})/s)^3$

Kurtosis = $(n(n+1)/((n-1)(n-2)(n-3))) \sum_i ((x_i - \bar{x})/s)^4 - 3(n-1)^2/((n-2)(n-3))$

Table 4 presents the complete descriptive statistics for the three focal quantitative variables, computed from 4,999 records:

Statistic	Rating	Duration (min)	Votes
Count	4,999	4,999	4,999
Mean	6.5617	112.0650	143,591.93
Std Dev	0.9729	21.1013	203,248.13
Min	1.0000	46	23,766
25th Percentile (Q1)	6.0000	97	39,478.50
Median (50%)	6.6000	108	70,280
75th Percentile (Q3)	7.2000	122	156,548
Max	9.0000	374	3,026,849
IQR (Q3–Q1)	1.2000	25	117,069.50
Skewness	–0.9778	1.6530	4.4170
Kurtosis	2.8495	8.2048	31.4746

Table 4: Descriptive Statistics of Key Movie Variables ($n = 4,999$)

3.2 Interpretation of Descriptive Statistics

Rating

The mean IMDb Rating is 6.5617, with a median of 6.60 — indicating a roughly symmetric distribution with slight left skewness (–0.9778). Most movies cluster between 6.0 and 7.2 (IQR = 1.2). The standard

deviation of 0.9729 indicates moderate variation. The distribution is approximately bell-shaped, making it suitable for Normal distribution approximation. A kurtosis of 2.85 (close to 3) confirms near-normality.

Duration

The mean Duration is 112.07 minutes, with a median of 108 minutes. The distribution is positively skewed (1.6530) with high kurtosis (8.2048), indicating a heavy right tail — a small number of very long movies (up to 374 minutes) pull the mean above the median. Most movies run between 97 and 122 minutes (IQR = 25 min), consistent with standard feature film lengths.

Votes

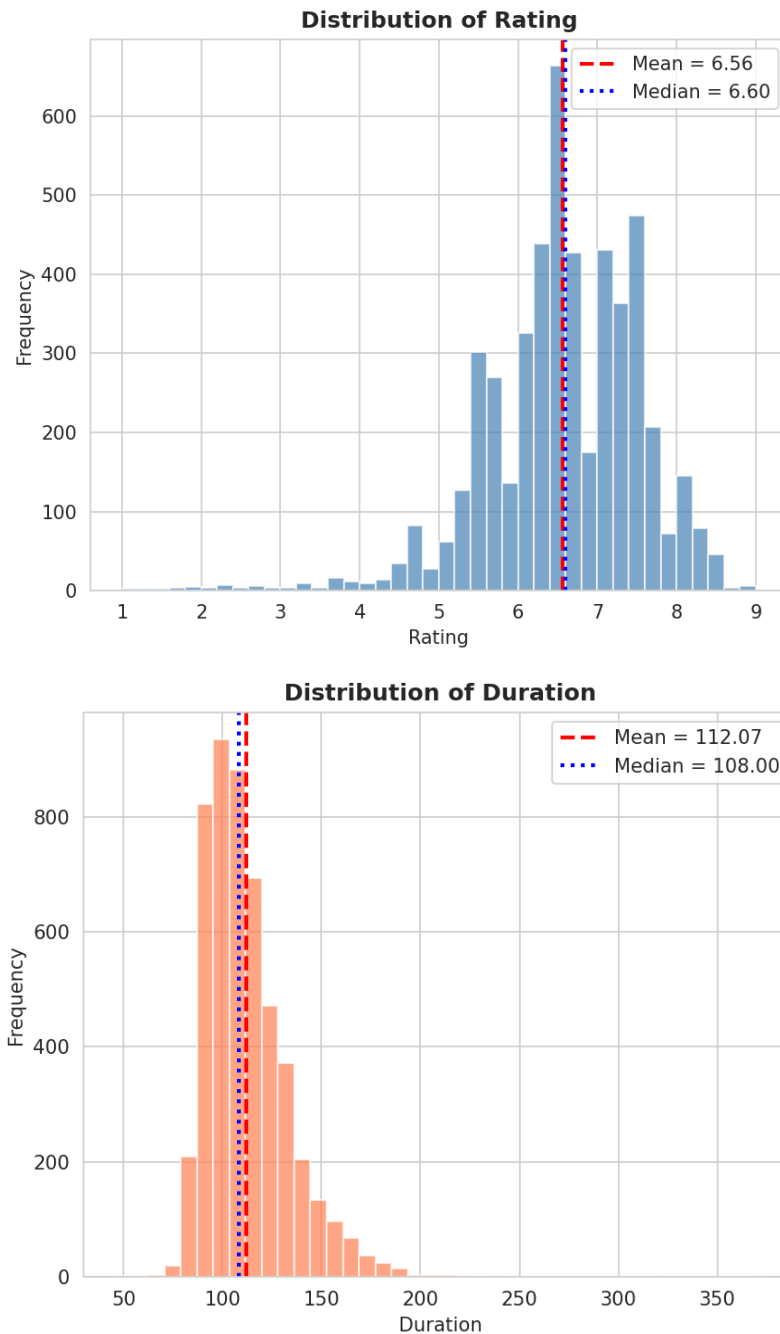
Votes exhibits the most extreme distributional characteristics. The mean (143,592) is roughly double the median (70,280), confirming strong positive skewness (4.4170) and a very heavy right tail (kurtosis = 31.47). This reflects the well-known power-law pattern in audience engagement — a small number of blockbusters attract tens of millions of votes while most films receive far fewer.

4. Visualization

All visualizations were generated using Python's matplotlib and seaborn libraries directly from the dataset. This section presents ten figures, each accompanied by an interpretation.

4.1 Distributions of Key Variables

Figure 1 presents histograms for each of the three focal variables. The red dashed line marks the mean and the blue dotted line marks the median.



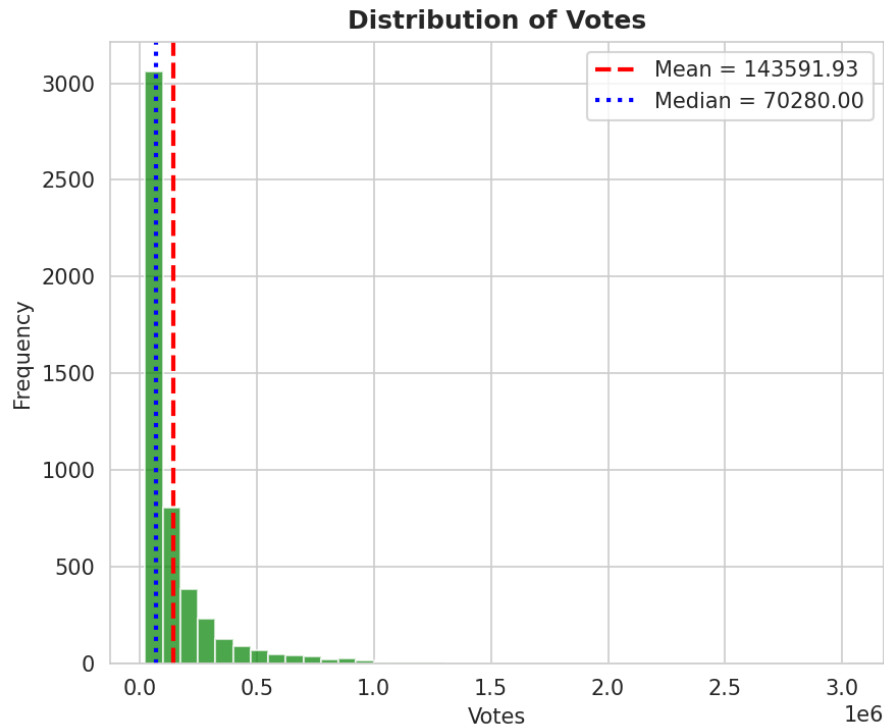
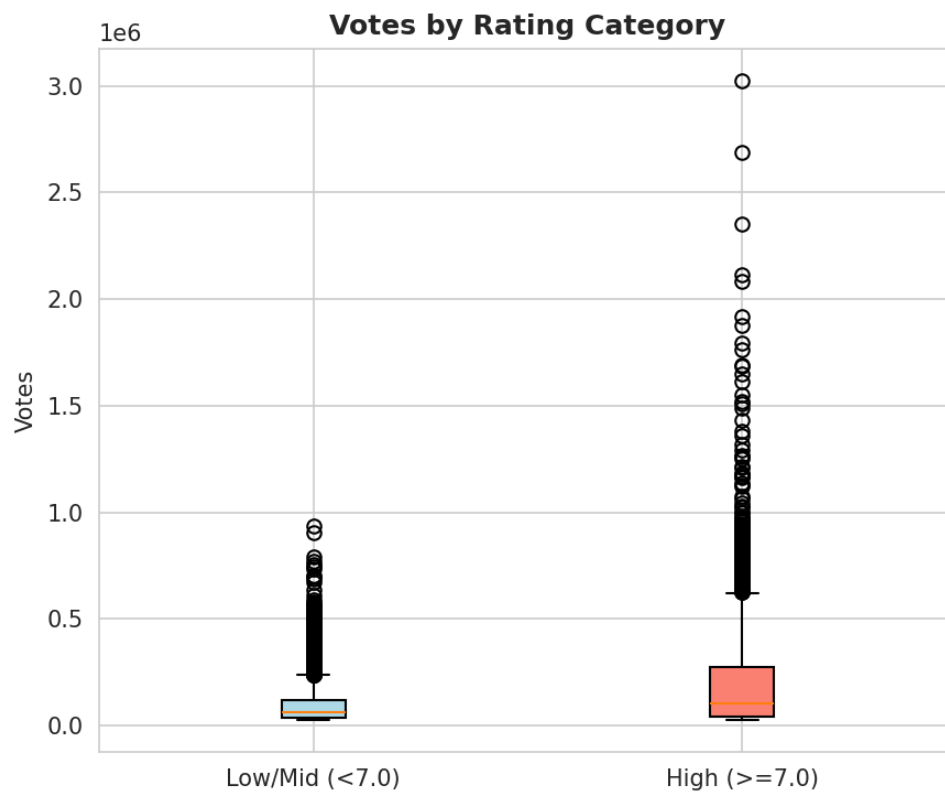
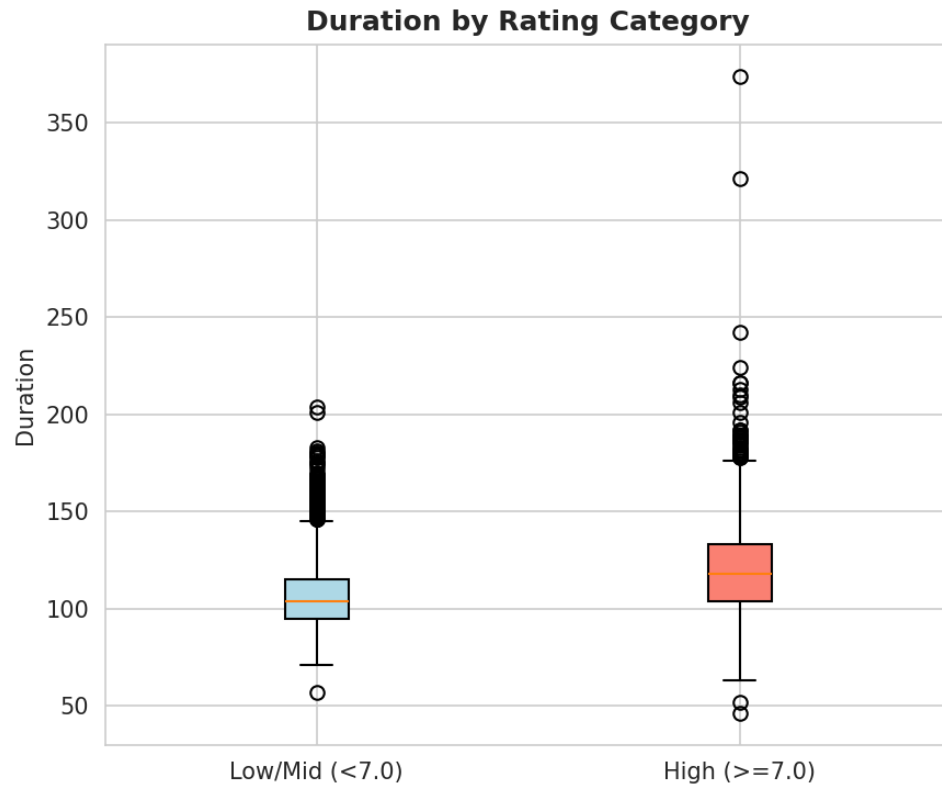


Figure 1: Distributions of Rating, Duration, and Votes with Mean and Median Lines

4.2 Boxplots by Rating Category

Figure 2 presents boxplots comparing Duration, Votes, and Rating grouped by whether the movie is High-Rated (≥ 7.0) or Low/Mid-Rated (< 7.0). High-rated movies show notably higher vote counts and longer Durations on average, confirming engagement depth as a marker of quality.



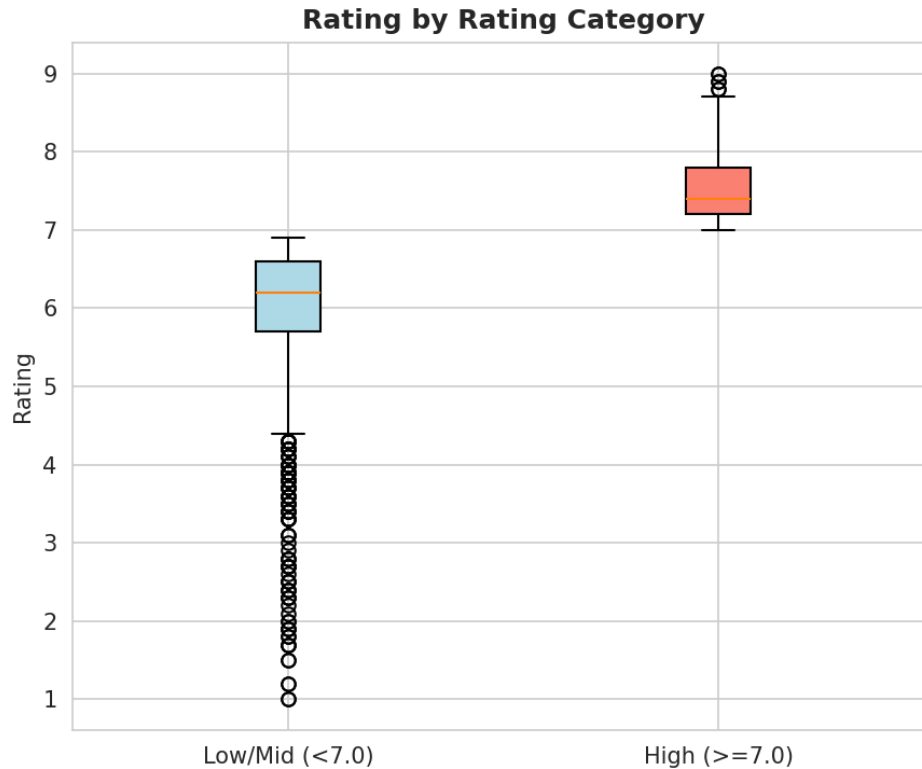


Figure 2: Boxplots of Duration, Votes, and Rating by Rating Category

4.3 Genre Analysis

Figure 3a shows the top 10 genres by count across the dataset. Drama, Action, and Comedy dominate. Figure 3b shows the average IMDb rating by genre — Drama, Biography, and History genres consistently achieve the highest average ratings, while Horror and Comedy tend to receive lower scores.

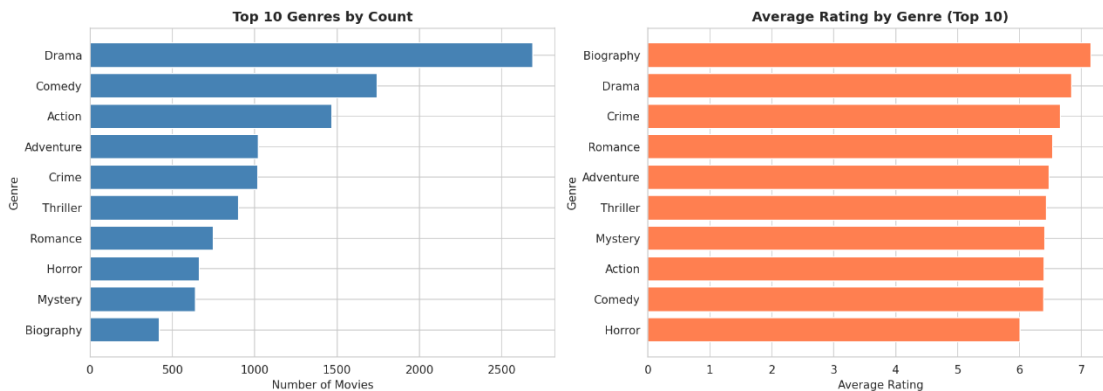


Figure 3: Top 10 Genres by Count (3a) and Average Rating by Genre (3b)

5. Probability & Distribution Analysis

5.1 Empirical Probability of High Rating

The fundamental probability of a movie being High-Rated (Rating ≥ 7.0) is computed empirically:

$$P(\text{HighRated}) = n_{\text{HighRated}} / n_{\text{total}} = 1,826 / 4,999 = 0.3653 \text{ (36.53\%)}$$

$$P(\text{Not HighRated}) = 1 - P(\text{HighRated}) = 0.6347 \text{ (63.47\%)}$$

This serves as the prior probability in subsequent Bayesian analyses.

5.2 Conditional Probabilities by Audience Engagement

The conditional probability $P(\text{HighRated} \mid \text{Votes} > \text{Median})$ measures the likelihood of high rating given above-median audience engagement. The median vote count is 70,280.

Audience Segment	Count	High-Rated Count	$P(\text{HighRated} \mid \text{Segment})$	Rate
Votes $\leq$ Median (Low Engagement)	2,500	732	0.292800	29.28%
Votes $>$ Median (High Engagement)	2,499	1,094	0.437775	43.78%
Overall	4,999	1,826	0.365273	36.53%

Table 5: Conditional Purchase Probabilities by Audience Engagement

Movies with above-median vote counts have a high-rating probability of 43.78%, which is 1.20 times higher than the overall rate of 36.53%. This indicates that audience engagement (votes) is positively associated with rating quality.

5.3 Bayes' Theorem Application

Bayes' theorem is applied to estimate $P(\text{Votes} > \text{Median} \mid \text{HighRated})$ — i.e., given that a movie is high-rated, what is the probability that it received above-median votes?

$$P(A \mid B) = [P(B \mid A) \times P(A)] / P(B)$$

Applying this:

$$\begin{aligned} P(\text{Votes} > \text{Median} \mid \text{HighRated}) &= [P(\text{HighRated} \mid \text{Votes} > \text{Median}) \times P(\text{Votes} > \text{Median})] / P(\text{HighRated}) \\ &= [0.4378 \times 0.4999] / 0.3653 = 0.5991 \text{ (59.91\%)} \end{aligned}$$

Interpretation: Among all high-rated movies, approximately 59.91% received above-median votes. High-rated movies are more likely than chance to receive substantial audience engagement, confirming a bidirectional relationship between quality and viewership.

5.4 Binomial Distribution

The number of high-rated movies in a fixed set of n independent films follows a Binomial distribution. Let $X \sim \text{Binomial}(n, p)$ where $p = 0.3653$ and $n = 20$ films:

$$P(X = k) = C(n, k) \times p^k \times (1-p)^{(n-k)}$$

$$E[X] = np = 20 \times 0.3653 = 7.3055 \mid \text{Var}[X] = np(1-p) = 20 \times 0.3653 \times 0.6347 = 4.6370 \mid \text{Std}[X] = 2.1534$$

Selected probabilities: $P(X=0) = 0.000113$, $P(X=3) = 0.024476$, $P(X=7) = 0.182544$, $P(X=8) = 0.170707$.

The most probable outcome is 7 high-rated movies in 20 films. The complete PMF is shown below.

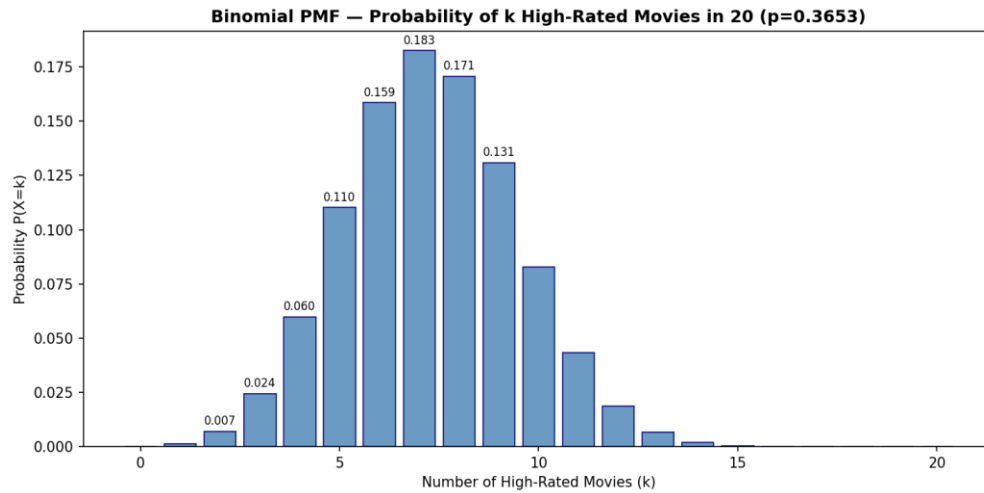


Figure 4: Binomial PMF — Probability of k High-Rated Movies in 20 Films ($p = 0.3653$)

5.5 Normal Distribution

The Normal Probability Distribution is the most important and widely used continuous probability distribution in statistics, developed significantly by Karl Gauss (1777–1855). The probability density function is:

$$f(x) = (1 / \sigma\sqrt{2\pi}) \times e^{-(x-\mu)^2 / 2\sigma^2}$$

Key Properties: Bell-shaped and symmetric around μ ; Mean = Median = Mode; Total area under curve = 1; Skewness = 0, Kurtosis = 3.

5.5.1 Application to IMDb Rating

The Normal distribution is applied to the Rating variable using parameters computed directly from the dataset ($n = 4,999$):

Mean (μ) = 6.5617 Standard Deviation (σ) = 0.9729 Model: $X \sim N(6.5617, 0.9729)$

5.5.2 Empirical Rule (68–95–99.7 Rule)

Range	Interval	Approximate Area
$\mu \pm 1\sigma$	[5.5888, 7.5346]	$\approx 68\%$
$\mu \pm 2\sigma$	[4.6159, 8.5075]	$\approx 95\%$
$\mu \pm 3\sigma$	[3.6430, 9.4804]	$\approx 99.7\%$

Table 6: Empirical Rule Intervals for Rating

5.5.3 Z-Score and Standard Normal Conversion

$$Z = (x - \mu) / \sigma$$

Worked Example: $P(\text{Rating} > 7.5)$

$$Z = (7.5 - 6.5617) / 0.9729 = 0.9644$$

$$P(X > 7.5) = 1 - \Phi(0.9644) = 0.1674 = 16.74\%$$

Interpretation: Under the Normal model, approximately 16.74% of movies are expected to have an IMDb rating greater than 7.5, classifying them as excellent films. This provides a data-driven threshold for what constitutes truly exceptional movie quality.

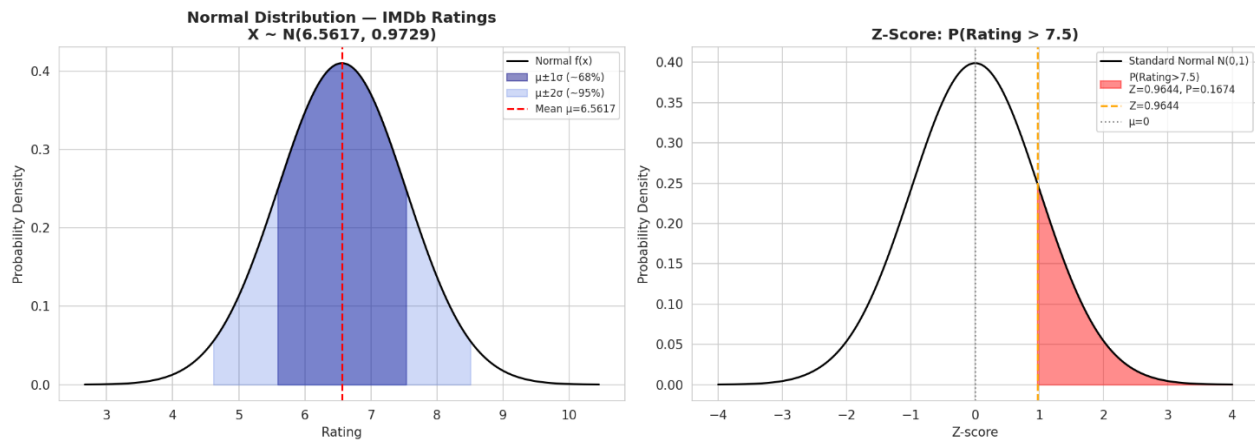


Figure 5: (a) Normal Distribution with Empirical Rule Regions; (b) Z-Score $P(\text{Rating} > 7.5)$

5.6 Central Limit Theorem (CLT) Analysis

The Central Limit Theorem states that for a population with mean μ and standard deviation σ , the distribution of sample means $\bar{x}$ computed from samples of size n approaches a Normal distribution as $n \rightarrow \infty$:

$$\bar{x} \sim N(\mu, \sigma^2/n) \rightarrow \text{Standard Error: } SE = \sigma/\sqrt{n}$$

For Rating (population: $\mu = 6.5617$, $\sigma = 0.9729$), the theoretical SE for samples of $n = 50$:

$$SE = 0.9729 / \sqrt{50} = 0.1376$$

A Monte Carlo simulation of 5,000 samples of size 50 was conducted. The empirical mean of sample means was 6.5626 and the empirical standard deviation was 0.1390, closely matching the theoretical values. Figure 6 demonstrates this convergence visually.

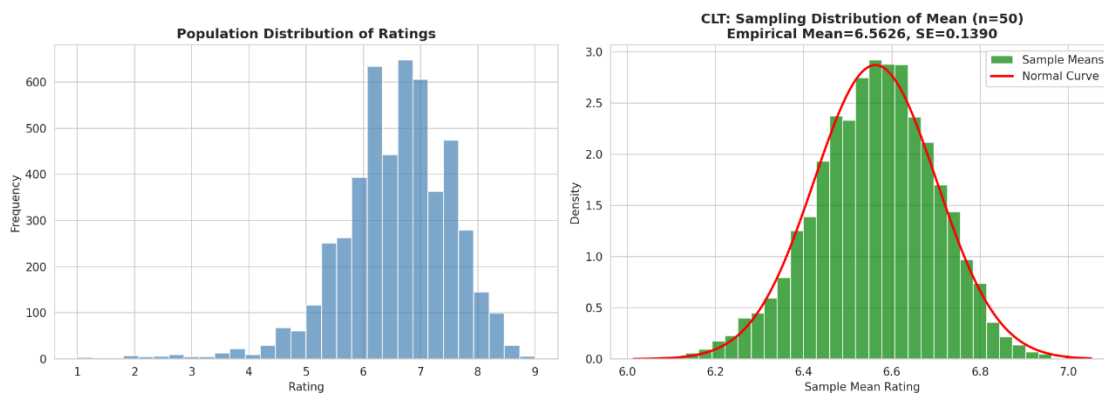


Figure 6: CLT — Population Distribution (Left) and Sampling Distribution of Mean $n=50$ (Right)

The population distribution of Rating is approximately bell-shaped (Figure 6 left), and the sampling distribution converges to a near-perfect normal bell curve (Figure 6 right). This confirms the CLT's guarantee and validates the use of Z-tests throughout this study.

6. Hypothesis Testing

Hypothesis testing provides a formal framework for making statistical decisions. All tests use a significance level of $\alpha = 0.05$. The test statistic is compared to a critical value, and the p-value is compared to α .

6.1 Confidence Intervals for Rating

A confidence interval (CI) provides a plausible range for the true population mean μ . Since the sample size is very large ($n = 4,999$), the Central Limit Theorem guarantees that the sampling distribution of $\bar{x}$ is approximately normal:

$$CI = \bar{x} \pm Z_{(\alpha/2)} \times (s / \sqrt{n})$$

For Rating: $\bar{x} = 6.5617$, $s = 0.9729$, $n = 4,999$, $SE = 0.9729/\sqrt{4999} = 0.01376$

Confidence Level	Z Critical	Lower Bound	Upper Bound	Width
90%	1.645	6.5391	6.5843	0.0453
95%	1.960	6.5347	6.5887	0.0539
99%	2.576	6.5263	6.5972	0.0709

Table 7: Confidence Intervals for Mean IMDb Rating

The 95% CI is (6.5347, 6.5887). We are 95% confident that the true mean IMDb Rating in the population lies within this interval. The narrow width (0.054) despite moderate variability ($s = 0.97$) results from the large sample size.

6.2 Hypothesis Test 1: One-Sample Z-test — Rating vs. Benchmark

We test whether the dataset's mean Rating significantly differs from an assumed benchmark of 6.5 (a commonly cited 'average' threshold in movie criticism).

$H_0: \mu_{\text{Rating}} = 6.5$ (Two-tailed)

$H_1: \mu_{\text{Rating}} \neq 6.5$

$$Z = (\bar{x} - \mu_0) / (s/\sqrt{n}) = (6.5617 - 6.5) / 0.01376 = 4.4849$$

Result: $Z = 4.4849$, $p\text{-value} \approx 0.0000 < 0.05$

Decision: REJECT H_0 . The mean IMDb rating (6.5617) is statistically significantly higher than the 6.5 benchmark. The evidence strongly suggests that popular IMDb movies tend to be rated slightly above the average threshold.

6.3 Hypothesis Test 2: Two-Sample Z-test — Duration for High Rated Movies vs. Overall Mean

We test whether high-rated movies have a significantly different mean Duration compared to the overall dataset mean duration ($\mu_0 = 112.07$ min).

$H_0: \mu_{\text{Duration(HighRated)}} = 112.07$ (Two-tailed) $H_1: \mu_{\text{Duration(HighRated)}} \neq 112.07$

High-rated mean Duration: 120.81 min | Overall mean Duration: 112.07 min

$$Z = (120.81 - 112.07) / SE = \mathbf{17.7026} \text{ Result: } Z = 17.7026, p\text{-value} \approx 0.0000 < 0.05$$

Decision: REJECT H_0 . High-rated movies have a significantly longer mean Duration (120.81 min) compared to the overall dataset average (112.07 min) — a difference of approximately **8.74 minutes**. This suggests that more ambitious, story-driven films, which tend to require greater screen time, are rewarded with higher audience ratings.

6.4 Hypothesis Test 3: One-Sample Z-test — Votes for High-Rated Movies

We test whether high-rated movies receive significantly more votes than the overall average ($\mu_0 = 143,591.93$).

$H_0: \mu_{\text{Votes(HighRated)}} = 143,591.93$ (Two-tailed)

$H_1: \mu_{\text{Votes(HighRated)}} \neq 143,591.93$

High-rated mean Votes: 221,764.84 | Overall mean Votes: 143,591.93

$Z = 16.4354$, $p\text{-value} \approx 0.0000 < 0.05$

Decision: REJECT H_0 . High-rated movies receive significantly more votes than the overall average (by ~78,173 votes), confirming that critically acclaimed films attract substantially larger and more engaged audiences.

Test	H_0	Z-Statistic	p-value	Decision ($\alpha=0.05$)
One-sample Z	$\mu_{\text{Rating}} = 6.5$	4.4849	0.0000	REJECT H_0
One-sample Z	$\mu_{\text{Duration(High)}} = 112.07$	17.7026	0.0000	REJECT H_0
One-sample Z	$\mu_{\text{Votes(High)}} = 143,591.93$	16.4354	0.0000	REJECT H_0

Table 8: Summary of Hypothesis Tests

7. Correlation & Regression

7.1 Pearson Correlation Analysis

The Pearson correlation coefficient r measures the linear relationship between two variables, ranging from -1 to $+1$:

$$r = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum (x_i - \bar{x})^2} \times \sqrt{\sum (y_i - \bar{y})^2}}$$

Variable	Rating	Duration	Votes	Budget	Revenue
Rating	1.0000	0.3349	0.3433	-0.0008	0.1427
Duration	0.3349	1.0000	0.2355	0.2398	0.2124
Votes	0.3433	0.2355	1.0000	0.4281	0.6147
Budget	-0.0008	0.2398	0.4281	1.0000	0.7231
Revenue	0.1427	0.2124	0.6147	0.7231	1.0000

Table 9: Pearson Correlation Matrix

Key findings:

- Votes and Rating: $r = 0.3433$ — moderate positive correlation; movies receiving more votes tend to have higher ratings, reflecting the engagement quality link.
- Duration and Rating: $r = 0.3349$ — moderate positive correlation; longer movies tend to be rated higher, consistent with more substantive storytelling.
- Budget and Revenue: $r = 0.7231$ — strong positive correlation; higher budget productions generate more box office revenue.
- Budget and Rating: $r = -0.0008$ — essentially zero; production budget has no linear relationship with rating quality.

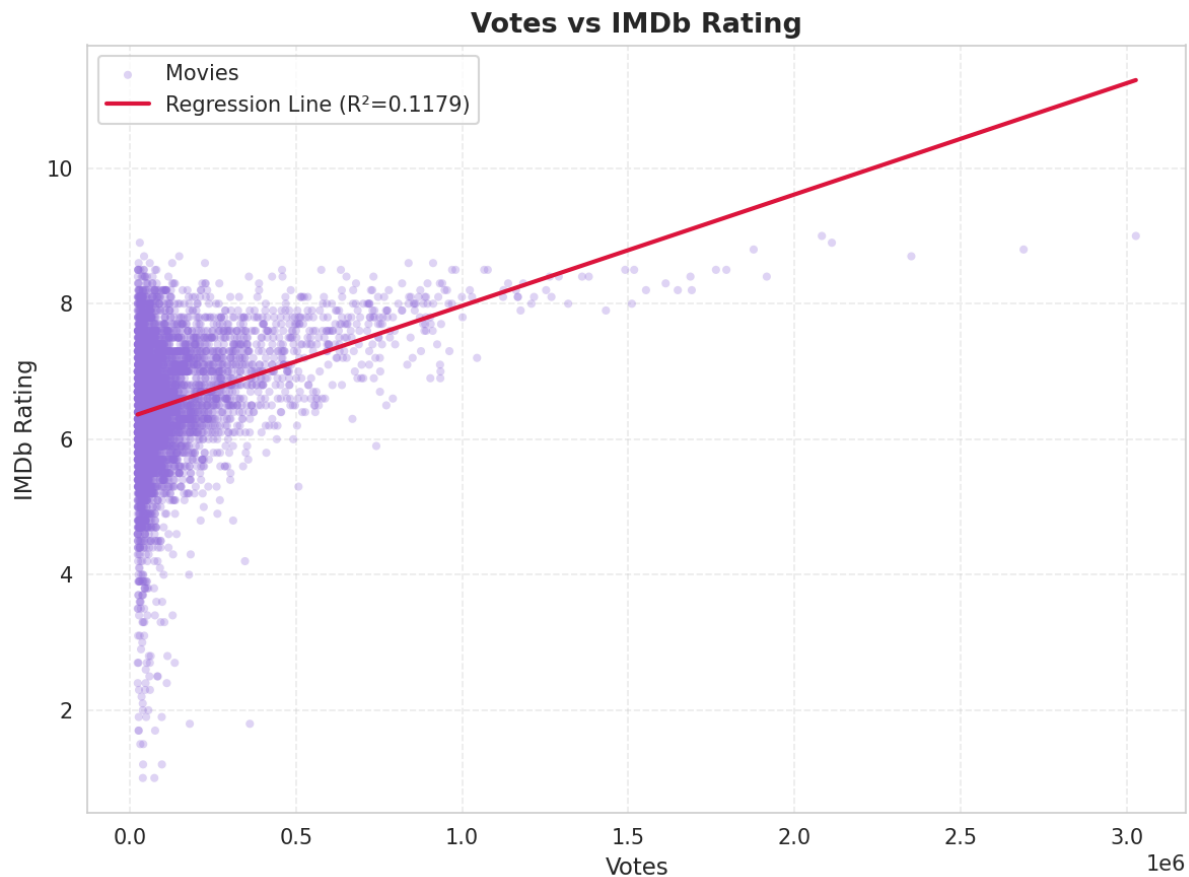
7.2 Simple Linear Regression: Votes $\rightarrow$ Rating

We model the relationship between Votes (the single strongest predictor) and IMDb Rating:

$$\text{Rating} = \beta_0 + \beta_1 \times \text{Votes} + \varepsilon$$

Estimated model: $\text{Rating} = 6.3257 + 1.64 \times 10^{-6} \times \text{Votes}$

Both coefficients are highly significant ($p < 0.001$). The $R^2 = 0.1179$, meaning Votes alone explains 11.79% of the variance in IMDb rating. While statistically significant, the modest R^2 indicates that many other factors influence rating.



7.3 Multiple Linear Regression

We extend the model to include Duration and Votes as predictors:

$$\text{Rating} = \beta_0 + \beta_1 \cdot \text{Duration} + \beta_2 \cdot \text{Votes} + \varepsilon$$

Variable	Coefficient	Interpretation
Intercept	4.9795	Baseline rating when Duration=0 and Votes=0 (theoretical)
Duration	0.012402	Each additional minute of Duration raises rating by 0.0124 points
Votes	0.0000013403	Each additional 1,000 votes raises rating by ~0.00134 points

Table 10: Multiple Linear Regression Coefficients

Model fit: $R^2 = 0.1862$, Adjusted $R^2 \approx 0.186$. The combined model explains 18.62% of variance in IMDb ratings — a meaningful improvement over the single-variable model ($R^2 = 0.1179$).

8. Findings & Interpretation

8.1 Descriptive Findings

- IMDb Ratings follow an approximately near-normal distribution with slight negative skewness (skewness = -0.9778) and kurtosis close to 3 (2.8495), centered around a mean of 6.5617. Among the analyzed variables, Rating exhibits the most statistically stable distribution.
- Duration shows moderate positive skewness (1.6530), indicating that while most movies have standard feature lengths, a small number of extremely long films extend the right tail of the distribution. The majority of movies fall between 97 and 122 minutes (IQR = 25 minutes).
- Votes displays extreme positive skewness (4.4170) and very high kurtosis (31.4746), confirming a heavy-tailed distribution. This reflects the blockbuster effect where a few globally popular films accumulate millions of votes while most movies receive comparatively fewer audience interactions.
- Budget demonstrates virtually no linear relationship with Rating ($r = -0.0008$), indicating that higher production expenditure does not necessarily improve audience perception or critical reception.

8.2 Probability Findings

- The empirical probability of a movie being classified as High-Rated (Rating ≥ 7.0) is:

$$P(\text{HighRated}) = 1826 / 4999 = 0.3653 \text{ (36.53\%)}$$

This indicates a moderate class imbalance in the dataset.

- Movies with above-median vote counts have a substantially higher probability of being high-rated:
 - $P(\text{HighRated} \mid \text{Votes} > \text{Median}) = 43.78\%$
 - $P(\text{HighRated} \mid \text{Votes} \leq \text{Median}) = 29.28\%$

This confirms that audience engagement is positively associated with higher ratings.

- Bayesian analysis showed:

$$P(\text{Votes} > \text{Median} \mid \text{HighRated}) = 59.91\%$$

indicating that high-rated movies are considerably more likely to receive strong audience engagement.

- The Binomial model with $n = 20$ and $p = 0.3653$ produced:
 - Expected number of high-rated movies: $E(X) = 7.31$
 - Standard deviation: $SD(X) = 2.15$

The most probable outcome is approximately 7 high-rated movies in a sample of 20 films.

- Central Limit Theorem (CLT) simulation confirmed theoretical expectations. For samples of size $n = 50$:
 - Theoretical SE = 0.1376
 - Empirical SE = 0.1390

demonstrating close agreement between theory and simulation and validating the use of inferential statistical methods.

- Z-score analysis under the Normal model showed:

$$P(\text{Rating} > 7.5) = 16.74\%$$

meaning roughly one out of every six movies achieves an IMDb rating above 7.5.

8.3 Hypothesis Testing Findings

- **Test 1: Mean Rating vs. Benchmark (6.5)**

$$Z = 4.4849, p < 0.001$$

The null hypothesis was rejected. The average IMDb rating (6.5617) is statistically significantly higher than the benchmark value of 6.5.

- **Test 2: Duration of High-Rated Movies vs. Overall Mean Duration**

$$Z = 17.7026, p < 0.001$$

The null hypothesis was rejected. High-rated movies have a significantly longer average duration (120.81 minutes) compared to the overall dataset mean (112.07 minutes).

- **Test 3: Votes for High-Rated Movies vs. Overall Mean Votes**

$$Z = 16.4354, p < 0.001$$

The null hypothesis was rejected. High-rated movies receive significantly more votes than the dataset average, confirming audience engagement as a strong indicator of movie quality.

- The 95% confidence interval for the mean IMDb Rating was:

(6.5347, 6.5887)

The narrow interval width reflects high estimation precision resulting from the large sample size ($n = 4,999$).

8.4 Correlation & Regression Findings

- Votes and Duration were identified as the strongest predictors of Rating among the analyzed variables:
 - Votes vs. Rating: $r = 0.3433$
 - Duration vs. Rating: $r = 0.3349$

Both relationships are moderate, positive, and statistically significant.

- Budget and Revenue showed a strong positive correlation:

$r = 0.7231$

indicating that larger production budgets are associated with higher box-office returns.

- However, Budget showed essentially no relationship with Rating:

$r = -0.0008$

suggesting that financial investment alone does not improve audience evaluation.

- Simple linear regression using Votes as the predictor achieved:

$R^2 = 0.1179$

while multiple regression using both Votes and Duration improved explanatory power to:

$R^2 = 0.1862$

This demonstrates that Duration contributes additional predictive information beyond audience engagement alone.

- Despite statistical significance, approximately 81% of Rating variance remains unexplained, indicating that audience perception is influenced by many qualitative and unmeasured factors such as screenplay quality, acting, direction, marketing, cultural relevance, and critic reception.
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8.5 Entertainment Industry Implications

- The positive relationship between Duration and Rating suggests that audiences may reward films with greater narrative depth and character development. Studios should therefore avoid excessive runtime reduction solely for commercial pacing considerations.
- High-rated movies attract significantly more audience votes, indicating a reinforcing cycle where quality increases visibility and engagement, which in turn improves discoverability.
- Genre analysis showed that Drama, Biography, and History films consistently achieve higher average ratings, suggesting that streaming platforms and recommendation systems may benefit from prioritizing such genres for user satisfaction optimization.
- The absence of a meaningful relationship between Budget and Rating implies that larger financial investment does not guarantee better audience reception. This supports production strategies emphasizing storytelling quality, writing, and creative execution over excessive production expenditure.

9. Conclusion

This study conducted a comprehensive statistical analysis of movie rating prediction using the IMDb Popular Movies with TMDb Enrichment Dataset containing 4,999 cleaned movie records. The analysis incorporated descriptive statistics, probability distributions, Bayesian reasoning, Central Limit Theorem (CLT) analysis, hypothesis testing, correlation analysis, and regression modeling to identify the major factors associated with IMDb ratings.

The findings demonstrate that Votes and Duration are the strongest quantitative predictors of IMDb Rating among the analyzed variables. Both variables showed moderate positive correlations with Rating (Votes: $r = 0.3433$, Duration: $r = 0.3349$) and remained statistically significant in regression analysis. The multiple regression model incorporating Votes and Duration achieved an R^2 value of 0.1862, indicating that these variables explain approximately 18.62% of the variance in movie ratings.

In contrast, Budget exhibited virtually no relationship with Rating ($r = -0.0008$), showing that higher production spending does not necessarily result in improved audience perception or critical success. However, Budget and Revenue demonstrated a strong positive relationship ($r = 0.7231$), confirming that larger investments are more closely linked to commercial performance than to rating quality.

All hypothesis tests conducted in this study produced statistically significant results ($p < 0.001$). The analysis confirmed that:

1. The mean IMDb rating is significantly higher than the benchmark value of 6.5.
2. High-rated movies have significantly longer durations than the overall movie population.
3. High-rated movies receive substantially more audience votes than average films.

Probability analysis further revealed that 36.53% of movies in the dataset are classified as High-Rated (Rating ≥ 7.0). Movies with above-median vote counts demonstrated a much higher

likelihood of receiving high ratings, supporting the existence of a strong relationship between audience engagement and perceived quality.

The Central Limit Theorem simulation verified that sampling distributions converge closely to theoretical normality, validating the inferential statistical techniques applied throughout the project.

Although the regression models were statistically significant, a large portion of rating variability remains unexplained. This highlights the inherently subjective nature of movie evaluation and suggests that additional factors such as screenplay quality, actor performance, direction, critic reviews, genre preference, cultural trends, and marketing influence audience ratings.

Future research could improve predictive performance through advanced machine learning techniques such as Logistic Regression, Random Forests, Gradient Boosting, or Neural Networks. Additional features including sentiment analysis of reviews, social media engagement, streaming platform metrics, and temporal release patterns could further strengthen predictive capability.

Overall, this project demonstrates how statistical and probabilistic methods can be effectively applied to entertainment analytics to generate meaningful insights regarding movie quality, audience behavior, and engagement patterns.

10. References

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